

Qns	Answer	Marks
9(a)(i)	Diffraction is the spreading of a wave into the geometric shadow when it passes through a slit or past an edge of an obstacle.	B1
9(a)(ii)	Wavelength of blue light is less than wavelength of red light Hence more orders seen OR each order is at a smaller angle than for the equivalent red	B1 B1 either statement
9(b)(i)	$d = \frac{\lambda}{2} = \frac{0.30}{2} = 0.15 \text{ m}$	A1
9(b)(ii)	N labelled at A and B, and nowhere else	B1
9(c)(i)	As M moves, path difference of between S ₁ M and S ₂ M changes continuously. When path difference between S ₁ M and S ₂ M is integer number of wavelength, constructive interference occurs, a loud sound is received by M When path difference between S ₁ M and S ₂ M off integer number of half wavelength, destructive interference occurs, a soft/no sound is received by M. <u>OR</u> The waves from S ₁ and S ₂ are of the same type, wavelength, frequency and amplitude. They travel opposite to each other, overlap and form a stationary wave, with displacement nodes and antinodes at regular intervals. M detects a large sound at displacement nodes, and soft (or no) sound at displacement antinodes.	B0 B1 B1 <u>OR</u> B1 B1
9(c)(ii)	$v = f\lambda$ $330 = 1100\lambda$ $\lambda = 0.30 \text{ m}$ distance between consecutive max (displacement nodes or pressure antinodes) = $0.30 / 2 = 0.15 \text{ m}$ No. of max M passes through in 1 s = $\frac{20}{0.15} = 133 = 130 \text{ (2 sf)}$ Hence frequency of the fluctuation of the sound is 130 Hz.	C1 wavelength C1 distance between max A1

9(d)(i)	This can be explained if the EM radiation is quantized, <u>with each photon having energy equal to hf</u>, h being the plank's constant and f being the frequency of the EM radiation. <u>Each electron absorbs energy from a single photon (hf).</u> <u>If the energy of photon is less than the work function of the metal ($hf < \Phi$), the electron will not be emitted</u>, no matter how many photons there are or how long the electron is exposed to the EM radiation.	B1 B1 B1
9(d)(ii)(1)	<u>Method 1</u> When $E_{\max} = 0$, $\frac{hc}{\lambda_0} = \Phi$. Hence, $\Phi = \frac{hc}{\lambda_0} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{300 \times 10^{-9}} = 6.63 \times 10^{-19} \text{ J} = 4.14 \text{ eV}$. <u>Method 2</u> Using $E_{\max} = \frac{hc}{\lambda} - \Phi$: As λ approaches infinity, $E_{\max}$ approaches Φ asymptotically. From Fig. 9.4, the (dotted) curve approaches -4 eV asymptotically. Hence, $\Phi \approx 4 \text{ eV}$.	M1 A1
9(b)(ii)(2)	$f_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{300 \times 10^{-9}} = 1.0 \times 10^{15} \text{ Hz}$	B1
9(b)(ii)(3)	According to Fig. 9.4, $E_{\max} = 20 \text{ eV}$ when $\lambda = 220 \text{ nm}$. Using $E_{\max} = eV_s$, or $V_s = \frac{E_{\max}}{e} = \frac{20 \text{ eV}}{e} = 20 \text{ V}$	M1 A1
9(c)	The same curve shifted downward (smaller x-intercept). Part of curve below the axis has to be dotted, same as the original curve. Curve must be reasonably smooth, and must not intersect with the original curve and the y-axis. The part of the curve below the horizontal axis must be dashed.	M1 A1 –1 for each mistake